Practice problems and exercises on Linear Algebra

February 7, 2026

Information

- There are two sections
  - Section A contains Problems along with solutions.
  - Section B contains exercises for you to try and solve.

Section A: Practice problems with solutions

Q1. Let set

$$S = \{(1, \frac{1}{2})', (\frac{1}{2}, 1)'\}$$

Check whether S is a linearly independent set.

Solution to Q1

By the definition of linear independence, if linear combination of the vectors of the set equated to zero vector implies that the coefficients are all equal to zero uniquely, then the set is linearly independent.

$$\alpha_1(1, \frac{1}{2})' + \alpha_2(\frac{1}{2}, 1)' = (0, 0)'$$

$$\implies (\alpha_1, \frac{\alpha_1}{2})' + (\frac{\alpha_2}{2}, \alpha_2)' = (0, 0)'$$

$$\implies (\alpha_1 + \frac{\alpha_2}{2}, \frac{\alpha_1}{2} + \alpha_2)' = (0, 0)'$$

$$\implies \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Check that $\det \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix} = \frac{3}{4} \neq 0$, so $\begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}$ is invertible. Hence,

$$\implies \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

So, S is linearly independent.

Q2. Let set

$$S = \{(1, \frac{1}{2})', (\frac{1}{2}, 1)', (0, 1)'\}$$

Check whether S is a linearly independent set.

Solution to Q2

By the definition of linear independence, if linear combination of the vectors of the set equated to zero vector implies that the coefficients are all equal to zero uniquely, then the set is linearly independent.

$$\alpha_1(1, \tfrac{1}{2})' + \alpha_2(\tfrac{1}{2}, 1)' + \alpha_3(0, 1)' = (0, 0)'$$

$$\implies (\alpha_1, \tfrac{\alpha_1}{2})' + (\tfrac{\alpha_2}{2}, \alpha_2)' + (0, \alpha_3)' = (0, 0)'$$

$$\implies \left(\alpha_1 + \tfrac{\alpha_2}{2},\ \tfrac{\alpha_1}{2} + \alpha_2 + \alpha_3\right)' = (0, 0)'$$

$$\implies \begin{pmatrix} 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

After substitution one will find that $(\alpha_1, -2\alpha_1, \frac{3\alpha_1}{2})'$ is solution. Hence, there are non-zero solutions. Therefore, S is linearly dependent.

Q3. Let **a** and **b** be two vectors given by

$$\mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

(a) Find the projection of **a** onto **b**.

(b) Find an orthonormal vector **v** orthogonal to **b**.

(c) Check whether $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$.

(d) Express **a** as a linear combination of **v** and **b**.

Solution to Q3

$$\textbf{Given: } \mathbf{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Projection of a onto b

The projection of $\mathbf{a}$ onto $\mathbf{b}$ is given by

$$\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a}^\top \mathbf{b}}{\mathbf{b}^\top \mathbf{b}} \mathbf{b}$$

$$\mathbf{a}^\top \mathbf{b} = (2, 1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 4, \quad \mathbf{b}^\top \mathbf{b} = (1, 2) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5$$

$$\operatorname{proj}_{\mathbf{b}} \mathbf{a} = \frac{4}{5} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ \frac{8}{5} \end{pmatrix}$$

Finding an orthonormal vector v orthogonal to b

A vector orthogonal to $\mathbf{b} = (1, 2)^\top$ is

$$\mathbf{u} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{since } \mathbf{b}^\top \mathbf{u} = 0$$

The norm of $\mathbf{u}$ is

$$\|\mathbf{u}\| = \sqrt{2^2 + (-1)^2} = \sqrt{5}$$

Hence the orthonormal vector is

$$\mathbf{v} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

Checking whether $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$

Consider a linear combination

$$\alpha \mathbf{v} + \beta \mathbf{b} = \mathbf{0}.$$

Taking dot product of both sides with $\mathbf{v}$, we get

$$\alpha\, \mathbf{v}^\top \mathbf{v} + \beta\, \mathbf{b}^\top \mathbf{v} = 0.$$

Since $\mathbf{v}$ is orthonormal and orthogonal to $\mathbf{b}$,

$$\mathbf{v}^\top \mathbf{v} = 1, \qquad \mathbf{b}^\top \mathbf{v} = 0.$$

Hence,

$$\alpha = 0.$$

Substituting $\alpha = 0$ in the original equation,

$$\beta \mathbf{b} = \mathbf{0}.$$

Since $\mathbf{b} \neq \mathbf{0}$, it follows that

$$\beta = 0.$$

Therefore, the vectors $\mathbf{v}$ and $\mathbf{b}$ are linearly independent, and hence $\{\mathbf{v}, \mathbf{b}\}$ is a basis for $\mathbb{R}^2$.

Expressing vector a as a linear combination of v and b
Since $\{\mathbf{v}, \mathbf{b}\}$ is an orthogonal basis,

$$\mathbf{a} = (\mathbf{a}^\top \mathbf{v})\mathbf{v} + \operatorname{proj}_{\mathbf{b}} \mathbf{a}$$

$$\mathbf{a}^\top \mathbf{v} = (2, 1)\frac{1}{\sqrt{5}}\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{3}{\sqrt{5}}$$

$$\mathbf{a} = \frac{3}{\sqrt{5}}\mathbf{v} + \frac{4}{5}\mathbf{b}$$

$$\boxed{\mathbf{a} = \frac{3}{\sqrt{5}}\mathbf{v} + \frac{4}{5}\mathbf{b}}$$

Q4. Find the rank of the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 1 & 2 & 1 \\ 4 & 5 & 6 \end{pmatrix}$$

and hence find the dimension of the null space of A.

Solution to Q4

Rank of a matrix is invariant to elementary row operations. Hence we apply the following row operations serially.

$$R_4 \leftarrow R_4 - 4R_1, \quad R_3 \leftarrow R_3 - R_1, \quad R_2 \leftarrow R_2 - 2R_1$$

$$\implies \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -4 \\ 0 & 0 & -2 \\ 0 & -3 & -6 \end{pmatrix}$$

Next, apply

$$R_4 \leftarrow R_4 - R_2$$

$$\implies \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -4 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{pmatrix}$$

Finally, apply

$$R_4 \leftarrow R_4 - R_3$$

$$\implies \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -4 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

The number of non-zero rows in the row-echelon form of A is 3.
Hence,

$$\text{rank}(A) = 3.$$

Dimension of the null space
The matrix A has 3 columns. By the Rank–Nullity Theorem,

$$\dim(\mathcal{N}(A)) = \text{number of columns} - \text{rank}(A) = 3 - 3 = 0.$$

$$\boxed{\text{rank}(A) = 3, \qquad \dim(\mathcal{N}(A)) = 0}$$

Q5. Find a basis for the row space of the matrix

$$A = \begin{pmatrix} 0 & 3 & 7 \\ 2 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}.$$

Solution to Q5

We find a row echelon form of A using elementary row operations.
First interchange R_1 and R_3:

$$R_1 \leftrightarrow R_3$$

$$\implies \begin{pmatrix} 1 & 2 & 4 \\ 2 & 1 & 1 \\ 0 & 3 & 7 \end{pmatrix}$$

Next,

$$R_2 \leftarrow R_2 - 2R_1$$

$$\implies \begin{pmatrix} 1 & 2 & 4 \\ 0 & -3 & -7 \\ 0 & 3 & 7 \end{pmatrix}$$

Next,

$$R_3 \leftarrow R_3 + R_2$$

$$\implies \begin{pmatrix} 1 & 2 & 4 \\ 0 & -3 & -7 \\ 0 & 0 & 0 \end{pmatrix}$$

The above matrix is in row echelon form.
Basis for the row space
The non-zero rows of the row echelon form form a basis for the row space of A.
Hence, a basis for the row space of A is

$$\{(1,\ 2,\ 4),\ (0,\ -3,\ -7)\}\,.$$

Section B: Exercises

The following problems are taken from *Higher Algebra*, S. K. Mapa, 12th Edition.

1. Show that the set S of vectors is linearly independent in $\mathbb{R}^3$.

$$\text{(i) } S = \{(1,2,3),(2,3,1),(3,1,2)\}$$

$$\text{(ii) } S = \{(0,1,1),(1,0,1),(1,1,0)\}$$

2. Show that the set S of vectors is linearly independent in $\mathbb{R}^4$.

$$\text{(i) } S = \{(1,2,3,0),(2,3,0,1),(3,0,1,2)\}$$

$$\text{(ii) } S = \{(1,1,1,0),(1,1,0,1),(1,0,1,1),(0,1,1,1)\}$$

3. Find the ranks of the following real matrices.

$$\text{(i)} \quad \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

$$\text{(ii)} \quad \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$$

$$\text{(iii)} \quad \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

4. Find a basis for the row space of the following matrices

$$\text{(i)} \quad \begin{pmatrix} 1 & 0 & 3 & 2 \\ 4 & 0 & 2 & 2 \\ 0 & 3 & 1 & 4 \end{pmatrix}.$$

$$\text{(ii)} \quad \begin{pmatrix} 2 & 1 & 3 & 5 \\ 3 & 4 & 1 & 2 \\ 0 & 3 & 1 & 1 \\ 5 & 5 & 4 & 7 \end{pmatrix}.$$